Week 6 – Quantum Computing

Teleportation, Deutsch–Jozsa, Bernstein–Vazirani

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Review from Week 5

- Last week, we studied a prototypical quantum algorithm in the *query model of computation*: the Deutsch algorithm.
- Given a quantum oracle O_f for a function $f : \{0,1\} \to \{0,1\}$ implementing

$$|x\rangle|y\rangle \longmapsto |x\rangle|y \oplus f(x)\rangle,$$

  we constructed a circuit that determines whether f is constant or balanced using a *single* query to O_f. Classically, two queries are required in the worst case.
- The speedup arises from quantum interference, and in particular from the phenomenon of *phase kickback*, which allows global properties of f to be encoded in relative phases.
- Today, we will see how this idea generalizes to functions with multiple input bits (the Deutsch–Jozsa algorithm), illustrating more clearly how quantum parallelism and interference combine to yield provable separations in the query model.
- We also examined nonlocal games, specifically the CHSH game, and saw how shared entanglement produces correlations that are strictly stronger than any classical strategy allows.
- In this lecture, we will see a different and perhaps more surprising application of entanglement: not to outperform classical correlations, but to enable the transfer of an *unknown quantum state* using only classical communication and a shared entangled pair. This protocol, known as *quantum teleportation*, highlights entanglement as a genuine informational resource.

Quantum Teleportation

The quantum circuit in the figure below executes the quantum teleportation protocol. In essence, it is a circuit that uses shared entanglement to transfer quantum information in the form of a (potentially unknown) qubit state using two classical bits of information. Suppose Alice & Bob share the entangled Bell state $|\Phi^+\rangle$, with Alice controlling the first wire, and Bob the second. Suppose also that Alice wishes to transfer a qubit $|\psi\rangle$ to Bob. Let us analyze how the circuit allows her to do so.

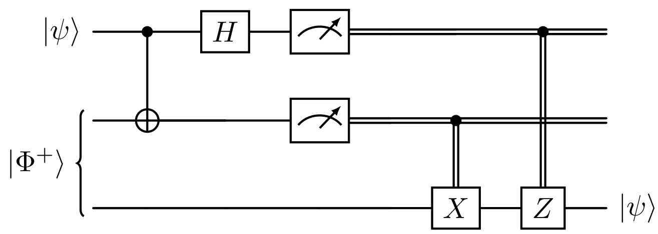


Figure 1: Quantum teleportation circuit.

Initial state. Let Alice's unknown qubit be $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$, with $|\alpha|^2 + |\beta|^2 = 1$.

Since Alice and Bob initially share the Bell state across the second and third wires, the initial three-qubit state is

$$|\psi\rangle \otimes |\Phi^+\rangle = (\alpha\,|0\rangle + \beta\,|1\rangle) \otimes \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\Big(\alpha\,|0\rangle\,|00\rangle + \alpha\,|0\rangle\,|11\rangle + \beta\,|1\rangle\,|00\rangle + \beta\,|1\rangle\,|11\rangle\Big).$$

Step 1: Alice applies a CNOT. Alice applies a CNOT with control on the first qubit and target on the second, yielding

$$\frac{1}{\sqrt{2}}\Big(\alpha\,|0\rangle\,|00\rangle + \alpha\,|0\rangle\,|11\rangle + \beta\,|1\rangle\,|10\rangle + \beta\,|1\rangle\,|01\rangle\Big).$$

Step 2: Alice applies a Hadamard on qubit 1. We obtain:

$$\frac{1}{\sqrt{2}}\Big(\alpha\,|+\rangle\,|00\rangle + \alpha\,|+\rangle\,|11\rangle + \beta\,|-\rangle\,|10\rangle + \beta\,|-\rangle\,|01\rangle\Big),$$

which, after regrouping the terms according to the first two qubits, simplifies to

$$\begin{aligned}|\Psi\rangle = \frac{1}{2}\Big[\,&|00\rangle\ (\alpha\,|0\rangle + \beta\,|1\rangle) + |01\rangle\ (\beta\,|0\rangle + \alpha\,|1\rangle)\\ &+ |10\rangle\ (\alpha\,|0\rangle - \beta\,|1\rangle) + |11\rangle\ (-\beta\,|0\rangle + \alpha\,|1\rangle)\Big].\end{aligned}$$

Note that the final ket in each parenthesis is the third qubit, the one that belongs to Bob.

Step 3: Alice measures the first two qubits. Measurement in the first two qubits produces two classical bits. Let the measurement outcome be $m_1 m_2 \in \{0,1\}^2$, where m_1 is the outcome from measuring the first qubit and m_2 from measuring the second.

From the expression above, each outcome occurs with probability 1/4, and Bob's post-measurement state becomes:

Alice's outcome ($m_1 m_2$)	Bob's state
00	$\alpha\,|0\rangle + \beta\,|1\rangle = |\psi\rangle$
01	$\beta\,|0\rangle + \alpha\,|1\rangle = X\,|\psi\rangle$
10	$\alpha\,|0\rangle - \beta\,|1\rangle = Z\,|\psi\rangle$
11	$-\beta\,|0\rangle + \alpha\,|1\rangle = XZ\,|\psi\rangle$

Step 4: Classical communication and Bob's correction. Alice sends the two classical bits (m_1, m_2) to Bob. Bob then applies the correction: $Z^{m_1} X^{m_2}$.

Exercise 1. *For each of the four cases $m_1 m_2 \in \{0,1\}^2$, show that Bob's qubit at the end of the teleportation protocol is the same as Alice's qubit at the start.*

Summary. The protocol transfers the unknown quantum state $|\psi\rangle$ from Alice to Bob using a shared entangled pair and two classical bits of communication. No physical particle carrying $|\psi\rangle$ is sent from Alice to Bob. Moreover, Bob cannot recover $|\psi\rangle$ until he receives Alice's classical bits; prior to that, his qubit contains no usable information about the state. Thus the protocol does not enable faster-than-light signaling. This is consistent with the *no-communication theorem*, which states that shared entanglement alone does not allow one party to transmit information instantaneously to another.

Deutsch–Jozsa Algorithm

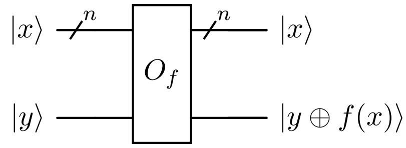
Just like the Deutsch algorithm studied last week, the Deutsch–Jozsa algorithm is another algorithm conceived and analyzed within the "query model of computation", with an added feature in the form of a "promise". More precisely, for $n \geq 1$, we are provided with a function $f : \{0,1\}^n \to \{0,1\}$ with the additional **promise** that this function is constant or balanced. In this setting, the latter means that exactly half of the inputs map to 0, and the other half map to 1. Notice a couple of things right away: for $n = 1$, this is exactly the problem solved by Deutsch algorithm. Secondly, note that when $n > 1$, there do exist functions that are neither balanced nor constant, hence the introduction of the additional **promise**. As before, the function is implemented as a black box, and the algorithm can query this black box with any valid input. The problem remains the same: what is the minimum number of queries needed to determine whether the function is constant or balanced?

Classical strategies – deterministic and randomized

1. It is clear that in the worst case, a deterministic strategy would require $2^{n-1} + 1$ queries. For instance, the first 2^{n-1} inputs used as queries could all map to a constant output, but one would need an additional query to determine if the function was constant or balanced.

2. However, one can do better with a randomized strategy. Suppose that we randomly sample k inputs (with replacement) to query f. If at least two of the inputs yield different outputs, we can conclude with certainty that the function is balanced. If each of the inputs yields the same output, we guess that the function is constant. Note that in this latter instance, if the function were actually balanced, the error probability, i.e. the probability of getting an identical output on each of the k inputs is $2\left(\frac{1}{2}\right)^k = 2^{1-k}$. For $k \approx 11$, this still indicates that our guess is corect with $\approx 99.9\%$ probability.

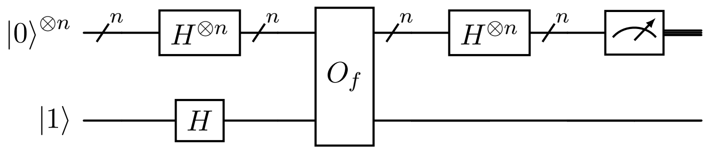
We will now see a quantum strategy that succeeds with a single query. As before, we need a unitary implementation of f, which works in the same way as before, except $|x\rangle$ is an n-bit binary string.



where $O_f : |x\rangle |y\rangle \mapsto |x\rangle |y \oplus f(x)\rangle$.

As before, the inner structure of this operator is unimportant, but note that $|y\rangle = |0\rangle$ as the second input yields $f(x)$ in the second output wire.

The complete Deutsch–Jozsa circuit is below.



Note first that the action of $H^{\otimes n}$ on $|0\rangle^{\otimes n}$ yields $|+\rangle^{\otimes n}$, which can be expanded using bilinearity as follows:

$$\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle.$$

Next, beginning with $|\psi_0\rangle = |0\rangle^{\otimes n}|1\rangle$, the state after applying the Hadamard gates on all $n+1$ qubits becomes

$$|\psi_1\rangle = \frac{1}{\sqrt{2^{n+1}}} \sum_{x\in\{0,1\}^n} |x\rangle\left(|0\rangle - |1\rangle\right).$$

Progressing through the circuit, we apply O_f on $|\psi_1\rangle$, and using the previously seen identity which states that for any bit $b \in \{0,1\}$, $|0\oplus b\rangle - |1\oplus b\rangle = (-1)^b\left(|0\rangle - |1\rangle\right)$, the resulting state transforms into:

$$|\psi_2\rangle = \frac{1}{\sqrt{2^{n+1}}} \sum_{x\in\{0,1\}^n} (-1)^{f(x)}|x\rangle\left(|0\rangle - |1\rangle\right).$$

At this point, we write $|\psi_2\rangle = |\psi_2'\rangle\,|-\rangle$, where

$$|\psi_2'\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{f(x)}|x\rangle.$$

The final step (before measurement) is to apply $H^{\otimes n}$ to $|\psi_2'\rangle$. Before we do that, let us formulate a compact expression to compute $H^{\otimes n}|x\rangle$, where $x \in \{0,1\}^n$. For a single bit $x_1 \in \{0,1\}$, we know that

$$H|x_1\rangle = \frac{1}{\sqrt{2}} \sum_{z_1\in\{0,1\}} (-1)^{x_1 z_1}|z_1\rangle.$$

(Verify that this is true by evaluating $H|0\rangle$ and $H|1\rangle$.)
Extending this to n bits, write $x = x_1x_2\cdots x_n$ and $z = z_1z_2\cdots z_n$. Then

$$\begin{aligned}
H^{\otimes n}|x\rangle &= \big(H|x_1\rangle\big)\otimes\cdots\otimes\big(H|x_n\rangle\big)\\
&= \left(\frac{1}{\sqrt{2}} \sum_{z_1\in\{0,1\}} (-1)^{x_1z_1}|z_1\rangle\right)\otimes\cdots\otimes\left(\frac{1}{\sqrt{2}} \sum_{z_n\in\{0,1\}} (-1)^{x_nz_n}|z_n\rangle\right)\\
&= \frac{1}{\sqrt{2^n}} \sum_{z\in\{0,1\}^n} (-1)^{x_1z_1}\cdots(-1)^{x_nz_n}|z\rangle\\
&= \frac{1}{\sqrt{2^n}} \sum_{z\in\{0,1\}^n} (-1)^{\sum_{i=1}^n x_iz_i}|z\rangle.
\end{aligned}$$

Define the (mod-2) inner product

$$x \cdot z \ := \sum_{i=1}^{n} x_i z_i \pmod 2,$$

so that the exponent above can be written as $x \cdot z$ (since only its parity matters). Thus we obtain the compact identity

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{x \cdot z} |z\rangle .$$

Now recall the state after the oracle call in the Deutsch–Jozsa algorithm (minus the final qubit $|-\rangle$):

$$|\psi_2'\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle .$$

Applying $H^{\otimes n}$ gives the state

$$\begin{aligned}
|\psi_3\rangle &= H^{\otimes n} |\psi_2'\rangle \\
&= \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} \left(H^{\otimes n} |x\rangle\right) \\
&= \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} \left(\frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{x \cdot z} |z\rangle \right) \\
&= \frac{1}{2^n} \sum_{z \in \{0,1\}^n} \left(\sum_{x \in \{0,1\}^n} (-1)^{f(x) + x \cdot z} \right) |z\rangle .
\end{aligned}$$

This is the final state of the algorithm immediately before measuring the first n qubits. The analysis now splits into two cases. In each case, we will measure all n qubits, and specifically focus on the amplitude (and as a result probability) associated with the potential outcome $|z\rangle = |0\rangle^{\otimes n}$. Note that in this case the (mod-2) inner product $x \cdot z = 0$.

Case 1: f is constant

Let $f(x) = c$ for all x; then the final state $|\psi_3\rangle$ becomes

$$|\psi_3\rangle = \frac{(-1)^c}{2^n} \sum_{z \in \{0,1\}^n} \left(\sum_{x \in \{0,1\}^n} (-1)^{x \cdot z} \right) |z\rangle$$

.
What this means is that the amplitude associated with the outcome $|z\rangle = |0\rangle^{\otimes n}$ (ignoring a potential global phase of $(-1)^c$) is

$$\frac{1}{2^n} \left(\sum_{x \in \{0,1\}^n} (-1)^{x \cdot z} \right) = \frac{1}{2^n} \left(\sum_{x \in \{0,1\}^n} (-1)^{0} \right) = \frac{1}{2^n} (2^n) = 1.$$

Exercise 2. *Show that when f is constant, the amplitude of any outcome $|z\rangle \neq |0\rangle^{\otimes n}$ is zero. That is, prove that*

$$\sum_{x \in \{0,1\}^n} (-1)^{x \cdot z} = 0 \quad \textit{for any } z \neq 0^n.$$

Case 2: f is balanced

Since f is balanced, we cannot factor $(-1)^{f(x)}$ out of the expression like in the constant case, but using $x \cdot z = 0$, the amplitude associated with the outcome $|z\rangle = |0\rangle^{\otimes n}$ becomes

$$\frac{1}{2^n}\left(\sum_{x\in\{0,1\}^n}(-1)^{f(x)}\right) = 0,$$

since exactly half of the $(-1)^{f(x)}$ terms evaluate to 1 and the other half evaluate to -1.

Exercise 3 (Bernstein–Vazirani Algorithm)**.** *Consider a following application of the Deutsch-Jozsa circuit. Note that the state $|\psi_3\rangle$ attained during the execution of the Deutsch Jozsa circuit (ignoring, as before, the $|-\rangle$ in the final register) is independent of the definition of f. Suppose the "promise" of the problem is that f is an* affine *function given by $f(x) = a \cdot x \oplus b$, where $a, x \in \{0,1\}^n$, $a \cdot x$ is the mod $-\ 2$ inner product, and $b \in \{0,1\}$. How would you determine a using a single query to the black box O_f implementing f?*